CH 2 Part 2

or the kernel

The nullspace of $N(L)$ is

$$N(L) = \{ x \in \mathcal{X} : Lx = 0 \}$$

XII Inner-sum and direct-sum spaces

Def 2.36: The inner sum space of two subspaces $V + W$ is

$$x = v + w \quad \forall \; v \in V \text{ and } w \in W$$

Def 2.3.7: Two linear subspaces are disjoint if

$$V \cap W = \{0\}$$

Lemma 2.3: V and W are subspaces of vector space S. $\forall \; x \in V + W$, there is a unique $v \in V$ and $w \in W$ s.t. $x = v + w$ iff V and W are disjoint.

Another way of combining subspaces is the direct sum

152 Def 2.38: The direct sum of two vector spaces, $V \oplus W$, is defined on the Cartesian product, so a point $V \oplus W$ is an ordered pair (v, w) with $v \in V$ and $w \in W$. Addition is defined component wise

$$(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2)$$

Scalar multiplication is defined as

$$\alpha(v, w) = (\alpha v, \alpha w)$$

VIII Direction of Vectors ; orthogonality

For vectors of $\mathbb{R}^3$ or $\mathbb{R}^2$

$$\cos\theta = \frac{\langle x, y\rangle}{\|x\|_2 \|y\|_2}$$

IX Weighted Inner Product

A weighted inner product can be obtained by inserting a Hermitian matrix, W, between the elements

$$\langle x, y\rangle_W = y^H W x$$

To use this as a norm we write

$$\|x\|_W^2 = \langle x, x\rangle_W = x^H W x$$

where W is a positive definite Hermitian matrix.

We can weight over integrals as well. Let $w(t) \geq 0 \ \forall [a,b]$, then we can write

$$\langle f, g\rangle_w = \int_a^b w(t) f(t) g(t)\, dt$$

A) Expectation as an Inner Product

Let X and Y be random variables

$$\langle X, Y\rangle = \int xy\, f(x) f(y)\, dx\, dy = E[XY]$$

X and Y are orthogonal if $E[XY] = 0$

The induced norm is written as

$$\langle x, x \rangle = E x^2$$

This can be expanded to a vector of random variables. Let $X = [X_1 \cdots X_n]^T$ and $Y = [Y_1 \cdots Y_n]^T$

$$\langle X, Y \rangle = E \sum_{i=1}^{n} X_i \overline{Y_i}$$

~~complex~~ conjugate

Thus the inner product

$$\langle Y, Y \rangle = E[Y^H Y] = tr(E[Y Y^H])$$

~~trace~~

IX Hilbert and Banach Spaces

Def 2.31: A complete normed vector space is a Banach space. A complete normed vector space with an inner product is called a Hilbert Space.

It can be shown that a finite dimensional normed vector space is complete. Hence every finite dimensional normed vector space is at least a Banach space.

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X Orthogonal Subspaces

Def 2.32: V and W are subspaces of S. V and W are orthogonal if

$$\forall\, v \in V,\ w \in W \quad \langle v, w \rangle = 0$$

Def 2.33: $V^{\perp}$ is the orthogonal compliment to V. It contains all the vectors orthogonal to V.

Theorem 2.5: Let $V, W < S$, where S is an inner product space, then.

1) $V^{\perp}$ is a closed subspace of S
2) $V \subset V^{\perp\perp}$
3) If $V < W$ then $W^{\perp} < V^{\perp}$
4) $V^{\perp\perp\perp} = V^{\perp}$
5) If $x \in V \cap V^{\perp}$ then $x = 0$
6) $\{0\}^{\perp} = S$ and $S^{\perp} = \{0\}$

XI Linear Transformations: Range and Nullspace

Def 2.34: A transformation $L: X \to Y$ is linear if $\forall\, x, x_1, x_2 \in X$

1) $L(\alpha x) = \alpha L(x) \quad \forall x \in X \ \& \ \alpha \in \mathbb{R}$
2) $L(x_1 + x_2) = L(x_1) + L(x_2)$

Def 2.35: $L: X \to Y$ is a linear operator, the range space is

$$R(L) = \{ y = Lx : x \in X \}$$

Theorem 2.6: V and W are linear subspaces of a linear space S. Then $V+W$ and $V \oplus W$ are isomorphic iff V and W are disjoint.

XIII Projections and Orthogonal Projections

Lets introduce the projection operator $P: S \to V$.
Then let $x = v + w$ then

$$Px = v$$

Def 2.39: A linear transformation P of a linear space is a projection if $P^2 = P$. (idempotent).

Def 2.40 Let P be a projection operator on the inner product space S. P is said to be an orthogonal projection if is range and nullspace are orthogonal $R(P) \perp N(P)$.

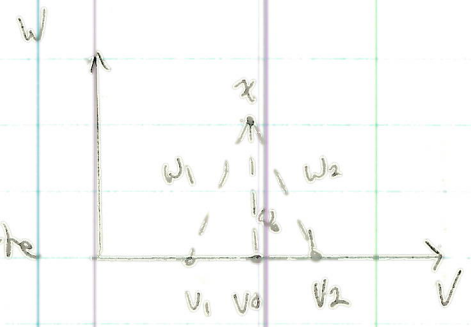
Theorem 2.7: Let P be a linear operator defined on a linear space S. Then the range and nullspace are disjoint linear subspaces of S and $S = R(P) + N(P)$. That is $R(P)$ and $N(P)$ are algebraic compliments.

poop fart :)

It is difficult to overstate the importance of projection as it is the key concept of most stochastic filtering and prediction theory in signal processing.

Consider the following problem. Given $x \in V \subset S$, what is the nearest point in V to x? Let

$$x = v_0 + w_0$$



From the figure, it is obviously w_0 which can be shown with the induced norm.

Now lets look at this in the light that v_0 is the nearest point to V and w_0 is the difference. Then rewriting the EQ above gives

$$w_0 = x - v_0$$

Which can be thought of as an error term. w_0 is minimized when the error is orthogonal to the data.

A) Projection Matrices

Let A be an $m \times n$ matrix

$$A = [P_1 \ P_2 \ P_3 \cdots P_n] \quad \text{s.t.} \quad P_n \text{ is } m \times 1$$

Then let the column space

$$V = \text{span}(P_1, P_2, \ldots, P_n) = R(A)$$

Theorem 2.8: Any (Hermitian) symmetric matrix with $P^2 = P$ is an orthogonal projection matrix

XIV The Projection Theorem

This section drives the point home that $v_0 \in V$ is the closest point to x if it is the orthogonal projection

Theorem 2.9: (Projection Theorem) Let S be a Hilbert space and let V be a closed subspace of S. $\forall x \in S$ $\exists$ $v_0 \in V$ that is unique and is the closest to x. That is

$$\| x - v_0 \| \leq \| x - v \| \quad \forall \ v \in V.$$

Furthermore v_0 is the minimizer of $\| x - v_0 \|$ iff $x - v \perp V$.

Theorem 2.10: Let V be a closed linear subspace of a Hilbert space S. Then

$$S = V \oplus V^{\perp}$$

and

$$V = V^{\perp\perp}$$

XI Orthogonalization of Vectors

It is often easier to compute over orthogonal basis vectors. It is usful then to be able to take any set of vectors T and produce an orthonormal set $T^{\perp}$ with the same span as T. This is what the Gram-Schmidt orthogonalization procedure does.

Given $T = \{p_1, p_2, \ldots, p_n\}$ we want to find a $T^{\perp} = \{q_1, q_2, \ldots, q_{n^{\perp}}\}$ s.t. $n^{\perp} \leq n$ and

$$\text{span}\{T^{\perp}\} = \text{span}\{T\}$$

and

$$\langle q_i, q_j \rangle = \delta_{ij}$$

The norm $\|\cdot\|$ in this section is the induced norm.

1) Normalize the vector

$$q_1 = \frac{p_1}{\|p_1\|}$$

2) Compute the difference between the projection q_2 and q_1.

$$e_2 = p_2 - \frac{\langle p_2, q_1 \rangle}{\|q_1\|^2} q_1 = p_2 - \langle p_2, q_1 \rangle q_1$$

if $e_2 = 0$ then $q_2 \in \text{span}(q_1)$ can be discarded.
If $e_2 \neq 0$ then

$$q_2 = \frac{e_2}{\|e_2\|}$$

3) A vector orthogonal to q_1 and q_2 obtained from the error between p_3 and its projection onto $\text{span}(q_1, q_2)$

$$e_3 = p_3 - \langle p_3, q_1 \rangle q_1 - \langle p_3, q_2 \rangle q_2$$

Normalize

$$q_3 = \frac{e_3}{\|e_3\|}$$

4) Generally the kth component is

$$e_k = p_k - \sum_{i=1}^{k-1} \langle p_k, q_i \rangle q_i$$

then normalize

$$q_k = \frac{e_k}{\|e_k\|}$$

A) A matrix-based implementation

This process can be written in matrix form.
Given $A = [p_1, \ldots, p_n]$ and the orthogonal vectors to be found $Q = [q_1, \ldots, q_n]$. Define R

$$R = \begin{bmatrix} \|p_1\| & \langle p_2, q_1 \rangle & \langle p_3, q_1 \rangle & \cdots & \langle p_n, q_1 \rangle \\ & \|p_2\| & \langle p_3, q_2 \rangle & \cdots & \langle p_n, q_2 \rangle \\ & & & & \vdots \\ & 0 & & & \|e_n\| \end{bmatrix}$$

XVI Some final Technicalities for Infinide-Dimensional Spaces.

The process shown was for finite-dimensional spaces. what happens if its infinite. Lets introduce a modified view of the basis.

A set T is said to be orthonormal if

$$\langle x_i, x_j \rangle = \delta_{i-j}$$

For an orthonormal set it can be shown that the infinite sum

$$\sum_{i=1}^{\infty} c_i P_i$$

converges iff $\sum_{i=1}^{n} \|c_i\|^2$ converges. Using what was shown above, the set of functions P_i are orthonormal and complete for a Hilbert Space.

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